

1 DATA AND REPRESENTATION

PhysioNet EEGMMIDB 104 subjects, 45 trials

BCI Competition IV 2a 9 subjects, 288 trials, 2 sessions

Cho2017 52 subjects, 200 trials

the same 8 sensorimotor channels, 8–30 Hz, 1 s window

spatial covariance
 $C \in \mathcal{S}_{++}^8$

density matrix
 $\rho = C/\text{tr } C$

an identity, not an analogy: 3 qubits, no dimensionality reduction, no angle embedding

2 FRAME

EEG nuisances (electrode gain, head geometry, skull conductivity, source mixing) act on C by congruence $C \mapsto ACA^\top$

sensor frame

$$\rho = C/\text{tr } C$$

invariant under $OCO^\top$ only,
the orthogonal subgroup

reference frame

$$\tilde{\rho} = WCW/\text{tr}(WCW)$$

$$W = M^{-1/2}$$

exactly invariant under $ACA^\top$,
 M label-free, training trials only

every suite is run in *both* frames, changing nothing else

3 PIPELINES

**classical
baselines**

CSP + LDA
TS + LR
MDM
FBCSP

**metric-matched
twin**

Riemannian
kernel SVM
log-Euclidean
as a second

**quantum
kernels**

HS overlap
Fidelity
HS-RBF
Bures-RBF
QRE-RBF

**circuit kernels
and ablation**

IQP entangled
IQP, entanglers
removed
PCA-matched

same covariances, same SVM, same tuning budget, same frame:
the *only* difference between these two is the metric

4 EVALUATION

nested cross-validation

5 folds $\times$ 3 repeats outer,
4-fold inner grid search
the identical search budget for every
pipeline, quantum and classical

paired per-subject statistics

Wilcoxon signed-rank, Holm corrected
within each pre-specified family
and two one-sided tests for
equivalence against the twin

settings: within-subject, cross-subject (leave one out),
cross-session, few-trial calibration, registers 3 to 6 qubits